

CS4865/6865

Assignment #1

Theoretical channel capacity

Due date: September 22, 2025

Instructions: To be completed individually. Format your submission as a single pdf file and submit on D2L by the due date.

Background reading:

1. WiFi standard 802.11b: https://en.wikipedia.org/wiki/IEEE_802.11b-1999
 2. B. Fadhah, A. Al-Lawati, S. Al-Maskari, A. Touzene, A. Al-Kindi, *Experimental performance evaluation of Wireless 802.11b networks*, 2008 First International Conference on the Applications of Digital Information and Web Technologies (ICADIWT), 2008.
- Login to the UNB Libraries server <https://lib.unb.ca/>
 - Click on the “Databases” menu.
 - Use “Browse Databases by Title” and select “IEEE Xplore Digital Library (Current & Archive)”
 - Search the IEEE database for the reference article (Note: to get a free access to the IEEE articles you need to search the database after logging in through the UNB Libraries web site).

Questions:

1. A typical 802.11b Wifi access point (AP) uses a 0.1W transmitter (other parameters, including the bandwidth and the carrier frequency, are listed in the references above). State the channel bandwidth and the carrier frequency used in 802.11b devices, and then plot the maximum theoretically possible data rate for the wireless channel from AP to a wifi user at distances from 10m to 100m. Use the isotropic antenna model for calculating the free space loss and assume the only interference is from the thermal noise with $T=400K$.
2. Compare your results from Question 1 to Figure 3 from A.B. Fadhah *et al* paper [2]. (Is the trend shown on the plots the similar? Are the measured values reported in the paper the same as the theoretical throughput values calculated in Question 2? Give a brief justification why they are (or are not) the same.)
3. Use Shannon’s channel capacity formula to calculate the theoretical maximum data rate for the direct-to-Earth (DTE) and direct-from-Earth (DFE) data links using the low gain antenna (LGA) installed on Curiosity rover, on Mars. Describe

the data communication hardware. Show your calculations. Provide the values of all parameters used in your calculations. Include references (where the specifications for the transmitters and receivers etc. can be found).

Hint: you may start your research with a wiki page on Mars exploration and then search for more detail about data communication with the rover, for example:

- “Communicating with Curiosity”

<https://science.nasa.gov/resource/communicating-with-curiosity-2/>

- “Talking to Martians” <https://sandilands.info/sgordon/communications-with-mars-curiosity>